System Explanation Notes

Face Verification Data Analysis · Geotagging Data Analysis · Attendance, Leave and Reports
Mobile Face Verification and Geotagged Attendance with Leave Management System — Universal Leaf Philippines Inc., Agoo, La Union

Part	Built with	Role
Mobile app	React Native + Expo	Captures the face photo and GPS location for each attendance scan
Web portal	React + Vite	Admin/HR and Supervisors review, monitor and analyse the data
Backend	NestJS + face-api.js	Checks location and face, then decides and saves the result
Database	PostgreSQL (Prisma)	Stores every scan's face and location data for analysis

> **Key rule:** the phone only gives early feedback. The **server re-checks location and face** before anything is saved.

1. Face Verification

1.1 How it works

Stage	What happens
Enrollment	HR registers the employee's face once (one face only, score ≥ 0.65, at least 120 px, no eyewear, consent given)
Liveness (blink)	Employee must blink once — stops printed photos and pictures on a screen
Lighting check	Face brightness must be ≥ 60 of 255, otherwise "add more light"
Auto-brighten	Dark photos are brightened before matching
Descriptor	The face is converted into 128 numbers
Matching	Live face's 128 numbers are compared with the enrolled face's

1.2 Formulas used

Measure	Formula	Meaning
Eye Aspect Ratio	(|p2−p6| + |p3−p5|) ÷ (2 × |p1−p4|)	Eye height ÷ width; drops when blinking
Blink	EAR < 97% of the person's normal EAR	Counts as one blink
Face brightness	0.299·R + 0.587·G + 0.114·B	Average brightness of the face area
Face distance	√Σ (enrolled − captured)^2 over 128 numbers	Smaller = more alike
Similarity score	(1 − distance ÷ 1.2) × 100	Shown as a percentage

1.3 Decision bands

Face distance	Similarity	Result
≤ 0.45	≥ 63%	**Approved — attendance saved**
0.45 – 0.60	50 – 62%	**Flagged — borderline, sent for review**
> 0.60	< 50%	**Flagged — mismatch, sent for review**
No face found	0%	**Rejected — retake in good lighting**

1.4 Face verification data collected

Data	Saved where	Used to analyse
Similarity score	Every approved and flagged scan	How strong each match was
Verification status	Approved / Pending review / Rejected / Archived	Approval and flag rates
Captured photo	Approved and flagged scans (with GPS stamp)	Visual proof for reviewers
Failure reason	Flagged scans	Borderline vs. mismatch
Mismatch / flag count	Per employee, per day	Repeated failed attempts
Reviewer decision	Reviewer, time, remarks	Review outcomes and speed
Face profile status	Active / archived per employee	Enrollment coverage

1.5 Face verification data analysis

Metric	How it is computed	What it tells you
Enrollment coverage	Employees with active face ÷ total employees × 100 (dashboard)	How many employees can use face attendance
Pending review	Count of flagged scans not yet reviewed (dashboard)	Reviewer workload right now
Approval rate	Approved scans ÷ (approved + flagged) × 100	How smoothly face matching works
Flag rate	Flagged scans ÷ (approved + flagged) × 100	How often faces don't match cleanly
Average similarity	Mean similarity score per employee or department	Who has weak enrolled photos or poor lighting
Repeated mismatches	Employees reaching 3 flags / mismatches in one day	Possible buddy-punching (someone else scanning)
Review outcome	Validated vs. rejected flagged scans	Real employee in bad light vs. impersonation

Pattern in the data	Likely meaning	Action
One employee always 50–62%	Poor enrolled photo	Re-register the face in good light
Many flags at one site / time	Dark area or backlight	Improve lighting at that spot
Flags mostly validated by reviewers	Genuine employees failing	Re-enroll, or enroll more samples
Flags rejected + 3 in a day	Someone else using the account	Supervisor/HR follow-up
Low enrollment coverage	Not everyone registered	Schedule face registration

Sample (illustrative week)	Calculation	Result
182 approved, 18 flagged	182 ÷ 200 × 100	**Approval rate 91%**
	18 ÷ 200 × 100	**Flag rate 9%**
Of 18 flagged: 15 validated, 3 rejected	15 ÷ 18 × 100	83% were the real employee
Enrollment: 95 of 100 employees	95 ÷ 100 × 100	95% coverage

1.6 Repeated failures

Event	What the system does
Each flagged scan	Saved to the review queue with photo, GPS and score (only the latest one per day stays)
3rd flag in a day	Supervisor (or HR) notified; employee notified; attendance on hold
Reviewer	Validate (becomes attendance) · Reject (hold lifted) · Archive
Next day	Counters reset to zero

2. Geotagging

2.1 How it works

Term	Meaning
Geotagging	Saving the location (coordinates, address, accuracy) with each scan
Geofence	A circle around a work site; attendance accepted only inside it
Geotagged area	A work site: center point, radius, allowed GPS accuracy, Office/Field type

Area setting	Example (Agoo plant)	Purpose
Center	16.358048, 120.353511	Middle of the site
Radius	1,000 m	How far from the center is accepted
Allowed accuracy	50 m (default)	Worst GPS accuracy accepted
Type	Office / Field	Office = one record per day; Field = several site visits

Step	Check	If it fails
1	GPS accuracy ≤ allowed accuracy	Rejected — "GPS accuracy is too low"
2	Distance to center ≤ radius	Rejected — "Outside the approved work location"
3	Only then is the face checked	Wrong place = face is never processed

2.2 Formula used (Haversine distance)

Step	Formula
A	$a = \sin^2(\Delta lat \div 2) + \cos(lat_1) \cdot \cos(lat_2) \cdot \sin^2(\Delta lon \div 2)$
B	$distance = 6{,}371{,}000\ m \times 2 \times atan2(\sqrt{a}, \sqrt{(1 - a)})$
Why	Gives the true distance on the Earth's curved surface, in meters

Scan	Accuracy	Distance	Result
1	±12 m	511 m	Accepted (inside)
2	±10 m	1,364 m	Rejected (outside)
3	±80 m	511 m	Rejected (weak GPS)

2.3 Geotagging data collected

Data	Description	Used to analyse
Latitude / Longitude	Exact position of the scan	Where attendance happened
Address	Readable address (OpenStreetMap)	Location shown in DTR and reports
GPS accuracy	Margin of error in meters	Signal quality per site
Distance from site	Meters from the area's center	Whether the radius fits the site
Work location	Which geotagged area was used	Site usage and field visits
Record type / visit no.	Office or Field, visit 1, 2, 3…	Field technician movement
Photo GPS stamp	Map, address, date and time on photo	Visual proof of location
Area assignments	Employees assigned to each area	Geotag coverage

2.4 Geotagging data analysis

Metric	How it is computed	What it tells you
Geotag coverage	Employees assigned to an area ÷ total employees × 100 (dashboard)	Who is set up for geotagged attendance
Geotagged logs today	Count of scans saved since midnight (dashboard)	Daily scan activity
Average distance	Mean distance from site per area	Where employees usually scan
Edge scans	Scans with distance close to the radius (e.g. > 80%)	Radius may be too small
Average GPS accuracy	Mean accuracy per area	Weak signal spots (indoors, buildings)
Scans per site	Count of scans grouped by work location	Most-used sites, field visit frequency
Field visits per day	Visits per field employee per day	Field technician productivity

Pattern in the data	Likely meaning	Action
Many scans near the radius edge	Circle too small for the site	Increase the radius
All scans very close to center	Circle larger than needed	Reduce the radius for tighter control
High GPS error at one site	Indoor / weak signal area	Scan near windows or outdoors; review allowed accuracy
Address differs from site name	Scan made at a different spot	Check the photo stamp
Low geotag coverage	Employees not assigned	Assign them to an area

Sample (illustrative, Agoo plant)	Calculation	Result
Distances: 120, 300, 450, 510, 870 m	(120+300+450+510+870) ÷ 5	Average 450 m
Edge scans (> 800 m of 1,000 m)	1 ÷ 5 × 100	20% near the edge
Accuracy: 8, 12, 15, 10, 20 m	65 ÷ 5	**Average ±13 m (good)**

Other detail	How it works
Phone pre-check	Checks location before opening the camera ("outside" or "weak signal")
Web browser	Accuracy limit relaxed to 2,000 m (no GPS chip); distance rule unchanged
Attendance mode	Assigned to a Field area → FIELD mode; Office areas only → FIXED (automatic)
Rejected scans	Outside / weak-GPS rejections are shown to the employee but **not saved**

3. Attendance Data Analysis

Measure	Formula
Late cutoff	Shift start + grace period
Late minutes	Arrival − late cutoff (0 if on time) → LATE if > 0, else PRESENT
Rendered Time In	Late arrivals round up to the next 30 minutes after shift start
Total worked	(Time Out − Rendered Time In) − break minutes
Undertime	(Shift end − undertime allowance) − Time Out (0 if not early)
Overtime	Time Out − shift end (0 if not past end)

Example (08:00–17:00)	Calculation	Result
Late	08:17 − (08:00 + 10 min grace)	**7 min, LATE**
Rendered Time In	08:17 rounded up	08:30
Total worked	(17:00 − 08:30) − 60 min break	7 h 30 m
Undertime / Overtime	Left at 17:00	0 / 0

No record on a day	Counted as
Approved leave	On Leave
Sunday / not a working day / before hire date	Not absent
Today, before late cutoff	Not absent yet
Otherwise	Absent

4. Dashboard and Reports

Indicator	How it is counted
Total employees	All employees except separated
Present / Late / Absent	Today's records + no-shows (field visits counted once per person)
Pending leaves / review	Pending leave requests / flagged face scans
Face enrollment	Enrolled ÷ total employees
Geotag coverage	Assigned to an area ÷ total employees
Calendar / Departments	Present, late, absent, on leave per day, week, month, department

Attendance rate	Rating
(Present + Late) ÷ (Present + Late + Absent + On Leave) × 100	**≥ 85% Good · 70–84% Warning · < 70% Poor**
Example: 18 Present, 3 Late, 2 Absent, 1 Leave	21 ÷ 24 = **88% Good**

Report	Contents
Attendance	Status counts; time in/out, total, late, undertime, overtime, location (exportable)
Leave	Requests per status; balances (remaining = earned − used)
Undertime filings	Employee reasons and review status
Filters	Date range and department (Supervisors see only their department)

5. Leave Management

Status	Meaning
Pending	Filed, waiting for review
Supervisor Approved	Approved by supervisor, waiting for HR
Approved	Final — days deducted from balance; employee can't Time In that day
Needs Revision / Rejected	Missing requirements / denied
Cancellation Pending / Cancelled	Cancel requested / cancel approved
Lapsed	Not decided in time, or employee worked all those days
Accrual	Leave credits added automatically every day at 1:00 AM when due

6. Security and Privacy

Control	How it works
Login and roles	One login; Admin/HR, Supervisor, Employee; Supervisors limited to their department
Server authority	Approval never trusts the phone's own face/location result
Face data	Processed on our own server; consent required; archived faces never shown
Audit log	Records area changes, face registration, every scan, flags and reviews

7. Limitations and Recommendations

Area	Limitation	Recommendation
GPS spoofing	Fake-GPS apps are not detected yet	Read the mock-location flag; flag impossible travel
Liveness	A video of the employee blinking could pass; server doesn't re-check the blink	Add a head-turn challenge; verify frames on the server
One face sample	Bad lighting causes extra flagged scans	Enroll 3–5 samples
Thresholds	0.45 / 0.60 are library defaults	Re-tune using pilot data (section 1.5)
Rejected scans	Outside / weak-GPS / no-face rejections aren't saved	Log them to analyse failed attempts
Photo storage	Photos stored in the database	Move to private file storage